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Chapter 1

Biological Foundation

This chapter is the biological foundation of the circuit that the model developed in this thesis follows: the neocortical microcircuit and the cortico-thalamo-cortical loop in which it is integrated. The account is canonical rather than exhaustive: it describes the organization that recurs across cortical areas rather than the full quantitative detail of any one, and it is drawn principally from sensory neocortex, where the underlying anatomy and physiology are most fully characterized. It rests on a small set of established sources: White [1] and the *Handbook of Brain Microcircuits* [2] for cortical structure, Douglas, Martin, and Whitteridge [3] and Douglas and Martin [4] for the canonical microcircuit, Rudy et al. [5] for the inhibitory cell classes, Sherman and Guillery [6, 7] for the thalamus and its cortical connections, and Thomson [8] for the internal organization of layer 6.

The chapter is deliberately confined to biology described on its own terms. It identifies the layers, cell types, and connections of the circuit, the direction and character of the information flow through it, and the distinct roles of its inhibitory and thalamic elements, but it does not map any of these onto components of the computational model. That correspondence is deferred to the bridge chapter. The separation is methodological, not presentational: the biology must stand as an independent description if the model is later to be evaluated against it, rather than being cast in whatever terms most flatter the model.

The unit of description is a single cortical column together with the thalamic circuitry to which it is reciprocally connected. The treatment stays at the level of cell classes and their connectivity; developmental origin, synaptic plasticity, and neuromodulation enter only where they bear directly on the circuit's operation, for instance in the control of thalamic firing mode. Where the canonical picture has since been refined or contested, this is noted rather than absorbed, so that the account remains anchored to the framework the rest of the thesis depends on.

The chapter proceeds from structure to circuit to loop: the laminar or-

ganization of the neocortex (Section 1.1), its excitatory principal neurons (Section 1.2) and inhibitory interneurons (Section 1.3), the canonical intracolumnar microcircuit that links them (Section 1.4), the thalamus and its relay and gating apparatus (Section 1.5), and the cortico-thalamo-cortical loop that binds cortex and thalamus into one processing system (Section 1.6), summarized in an integrative circuit diagram (Section 1.7).

1.1 Laminar organization of the neocortex

The defining structural feature of the neocortex is its organization into horizontal layers, or laminae, stacked between the pial surface and the underlying white matter. The layers are distinguished cytoarchitecturally, by the size, packing density, and type of their constituent neurons, rather than by sharp anatomical borders, and the conventional scheme recognizes six of them, numbered L1 to L6 from the pia inward [1, 2]. This laminar scheme is most sharply expressed in primary sensory cortex, the granular or koniocortical type, in which the input layer is especially prominent; in agranular cortex, such as primary motor cortex, that layer is thin or absent. The canonical account developed here, and the sources it rests on, are drawn principally from granular sensory cortex, where the layers and their connections are most fully characterized.

The superficial layers occupy the outermost cortex. Layer 1 is a cell-sparse zone dominated not by somata but by neuropil: the apical dendritic tufts of pyramidal cells whose bodies lie in deeper layers, together with horizontally running axons, among them the terminals of long-range feedback projections from higher cortical areas. It contains only a sparse population of inhibitory neurons and no excitatory somata to speak of, and functionally it is best understood as the principal termination zone for top-down input onto the distal dendrites of pyramidal cells [2]. Beneath it lie layers 2 and 3, frequently treated together as the supragranular compartment. They are populated by small and medium pyramidal cells and form the associative tier of the cortex; they are the main origin of the feedforward corticocortical projections that carry a cortical area's output to the next area in the hierarchy [4].

Layer 4 is the granular input layer. It is densely packed with small excitatory neurons, and it is the principal cortical target of the ascending thalamocortical afferents that convey sensory information to the cortex [1, 4]. Its prominence is what distinguishes granular sensory cortex, and its role as the point of entry for thalamic input makes it the anchor of the layered scheme: layers are conventionally grouped by their position relative to it, as supragranular (L1–L3), granular (L4), and infragranular (L5–L6).

The deep, infragranular layers are the cortex's principal output stage. Layer 5 contains the largest pyramidal cells in the cortex and is the major

source of long-range projections to subcortical targets; it also projects to the thalamus, in a manner distinct from that of layer 6 and taken up in Section 1.6. Layer 6 is the origin of the numerically dominant descending projection back to the thalamus, and it also sends axons locally upward toward layer 4 [2, 6]. The corticothalamic cells that give rise to this projection are, however, only part of the layer: they make up some 30–50% of its pyramidal cells, and the remainder are corticocortical cells whose axons stay within the cortex [8]. The two populations and their inputs are described in Section 1.4. Together the deep layers return the results of cortical processing to subcortical structures and, critically for this thesis, close the loop with the thalamus.

This horizontal lamination is only one axis of cortical organization. Cutting across it is the vertical, or radial, dimension: the apical dendrites of pyramidal cells and the predominantly vertical interlaminar axons bind neurons across all six layers into the columnar unit that was taken as the object of description at the outset of this chapter. Lamination specifies what each layer is and where it sits; the directed flow of activity between the layers, which constitutes the canonical microcircuit, is the subject of Section 1.4, after the cell types that populate the layers have been introduced in Sections 1.2 and 1.3.

1.2 Excitatory principal neurons

The large majority of cortical neurons, on the order of 80%, are excitatory, glutamatergic cells, and the principal type among them is the pyramidal cell [1, 2]. Its morphology is stereotyped and functionally consequential: a roughly pyramid-shaped soma gives rise to a skirt of basal dendrites near the cell body, a set of oblique dendrites, and a single prominent apical dendrite that ascends toward the pial surface and terminates in a branching tuft in layer 1. The dendrites are densely studded with spines, the principal sites of excitatory input, and the axon projects both locally, within and across layers of the same column, and, for many pyramidal cells, over long range to other cortical areas or subcortical targets. Pyramidal cells are found in every layer except layer 1, and it is the layer of the soma that largely determines a cell's projection target, as set out below.

The pyramidal cell's dendritic geometry segregates its inputs into two functionally distinct streams. Feedforward input, which ascends from lower cortical areas and, in the input layer, from the sensory thalamus, arrives predominantly on the basal and perisomatic dendrites, electrotonically close to the soma and the site of action-potential initiation. Feedback input from higher cortical areas arrives instead on the apical tuft in layer 1, electrotonically remote from the soma. Because the apical tuft is separated from the perisomatic region by the long apical trunk, the two streams are not simply

summed: the distal compartment supports its own regenerative events, and near-coincident activation of the perisomatic and apical compartments can trigger dendritic calcium spikes that switch the cell into burst firing [9]. The pyramidal cell is therefore not a simple integrator but a two-compartment device that can register the coincidence of bottom-up and top-down signals, a property whose circuit context is developed in Sections 1.4 and 1.6.

The projection target of a pyramidal cell is characteristic of its layer. Cells in layers 2 and 3 are the principal source of feedforward corticocortical output, projecting to the next area in the cortical hierarchy while also contributing dense recurrent connections within the local circuit [4]. Layer 5 contains two broad classes: intratelencephalic cells, which project to other cortical areas and to the striatum, and the large, thick-tufted pyramidal-tract cells, which project to subcerebral structures and, through axon collaterals, supply driving input to higher-order thalamic nuclei [2, 6]. Cells in layer 6 include the corticothalamic pyramidal cells that form the numerically dominant descending projection to the thalamus, alongside cells projecting corticocortically and axons ascending locally to layer 4 [2]. The distinct thalamic roles of the layer-5 and layer-6 projections are the subject of Section 1.6.

Not all excitatory neurons are pyramidal. Layer 4 contains, in addition to small pyramidal cells, a population of spiny stellate cells: excitatory, spine-bearing neurons that lack a prominent apical dendrite and whose dendrites instead radiate locally around the soma [1, 4]. They are a specialization of the input layer, receiving thalamocortical afferents directly and distributing that input locally within the column, and they are most numerous in the primary sensory areas where the granular layer is best developed. Functionally they can be regarded as the first cortical recipients of the ascending sensory message, a role taken up in the description of the canonical microcircuit in Section 1.4.

With the excitatory principal neurons and their two input streams established, the remaining cellular component of the column is the inhibitory interneuron population, which is small in number but decisive in shaping how the excitatory cells integrate these streams. It is described next, in Section 1.3.

1.3 Inhibitory interneurons

The remaining fifth or so of cortical neurons are inhibitory, GABAergic interneurons. Though a numerical minority, they are not a homogeneous population: they fall into a small number of largely non-overlapping classes, each distinguished by the molecular markers it expresses, the firing pattern it exhibits, and, most consequentially for the circuit, the subcellular compartment of the pyramidal cell it innervates [5, 10]. This last property is the organizing principle of this section: the principal interneuron classes di-

vide the pyramidal cell between them, one governing the perisomatic region where output is generated and another governing the distal dendrites where the two input streams of Section 1.2 are received, so that inhibition is not a single global quantity but a compartment-specific control.

Three molecularly defined classes account for nearly the entire interneuron population: cells expressing parvalbumin (PV, roughly 40%), cells expressing somatostatin (SST, roughly 30%), and cells expressing the ionotropic serotonin receptor 5HT3aR (roughly 30%) [5]. The three classes are essentially non-overlapping, and together they exhaust the inhibitory population to a first approximation. The four components named in the scope of this chapter (PV, SST, VIP, and the interneurons of layer 1) map onto this scheme as follows: PV and SST are the first two classes, while VIP cells and the layer-1 interneurons are two functionally distinct subgroups of the third, the 5HT3aR class, as set out below.

PV cells are fast-spiking and target the perisomatic region of the pyramidal cell: the two morphological types are the basket cells, which innervate the soma and proximal dendrites, and the chandelier, or axo-axonic, cells, which innervate the axon initial segment itself [10]. Because they act at or near the site of action-potential initiation, PV cells are positioned to control the pyramidal cell's spike output directly, that is, its gain and its timing, rather than the integration of its inputs; their fast kinetics and dense, relatively non-selective local connectivity also make them the principal substrate of fast network synchronization in the gamma band [10]. SST cells, by contrast, target the distal dendrites. The archetype is the Martinotti cell, whose ascending axon ramifies in layer 1 among the apical tufts of pyramidal cells [10]; SST inhibition therefore acts on the compartment that receives top-down feedback, and is positioned to control dendritic integration and the regenerative dendritic events described in Section 1.2 rather than somatic output.

The 5HT3aR class contains the remaining two components. VIP cells, expressing vasoactive intestinal peptide, are the disinhibitory element of the circuit: rather than targeting pyramidal cells directly, they preferentially inhibit other interneurons (SST cells above all, and PV cells to a lesser degree), so that their activation releases the pyramidal cell, and specifically its dendrites, from inhibition [10]. This VIP→SST→pyramidal motif is the canonical disinhibitory pathway by which long-range and neuromodulatory inputs, which contact VIP cells, can open the dendritic compartment to feedback. The layer-1 interneurons form the other, non-VIP part of the 5HT3aR class; the most abundant type is the neurogliaform cell, which releases GABA diffusely by volume transmission rather than at conventional one-to-one synapses, producing slow, spatially broad inhibition (including through $GABA_B$ receptors) onto the apical tufts in layer 1 [10]. Functionally they are the inhibitory counterpart of the feedback that terminates in layer 1. The resulting division of the pyramidal cell among the four inhibitory

components is summarized in Figure 1.1.

A caution is needed on what this compartmental division of labour does and does not establish. The synaptic targeting itself, PV to the perisomatic region and SST to the distal dendrites, is robust and is the property this chapter relies on. The further step of assigning each class a fixed arithmetic operation on the pyramidal cell's input-output function, PV acting divisively and SST subtractively, is often made but is not settled: activating either class has been shown to produce mixtures of divisive and subtractive effects, and threshold nonlinearities interacting with recurrent network activity can convert subtractive inhibition of single neurons into divisive inhibition at the network level, or the reverse [11]. This chapter therefore characterizes the interneuron classes by their synaptic targeting and their consequent control of the output versus the input compartment, and does not commit to a fixed arithmetic label; the computational interpretation of these operations is taken up where the model is introduced.

With the excitatory and inhibitory cell types in hand, the layers they populate can now be connected into a directed circuit. Section 1.4 assembles the canonical intracolumnar microcircuit, and the interneuron classes described here reappear there as the compartment-specific controllers of its excitatory flow.

1.4 The canonical intracolumnar microcircuit

The layers and cell types described so far are bound into a directed circuit whose principal features recur across cortical areas and species. This recurring pattern of interlaminar connectivity is the canonical microcircuit, abstracted by Douglas, Martin, and Whitteridge [3] and developed by Douglas and Martin [4]: despite the areal diversity of the cortex, the dominant excitatory pathways between the layers follow a stereotyped template, which serves here as the organizing description of intracolumnar flow.

The template has a characteristic feedforward spine. Ascending thalamocortical afferents deliver their input principally to layer 4, whose excitatory neurons project upward to the supragranular layers 2 and 3. Layers 2/3 in turn project to layer 5, and layer 5 to layer 6, so that the dominant axonal pathway runs thalamus $\rightarrow$ L4 $\rightarrow$ L2/3 $\rightarrow$ L5 $\rightarrow$ L6 [4, 1]. The infragranular layers constitute the circuit's output stage, with layer 5 projecting to subcortical and higher-order thalamic targets and layer 6 back to the thalamus (Section 1.6), while layer 6 also sends axons locally upward to layer 4, providing one intracolumnar feedback limb of the loop [2]. This serial spine, from granular input through supragranular association to infragranular output, is the sense in which the column "processes" its input.

The final step of this spine, from layer 5 to layer 6, is of a different kind from the steps that precede it. The projections from layer 4 to layers 2/3,

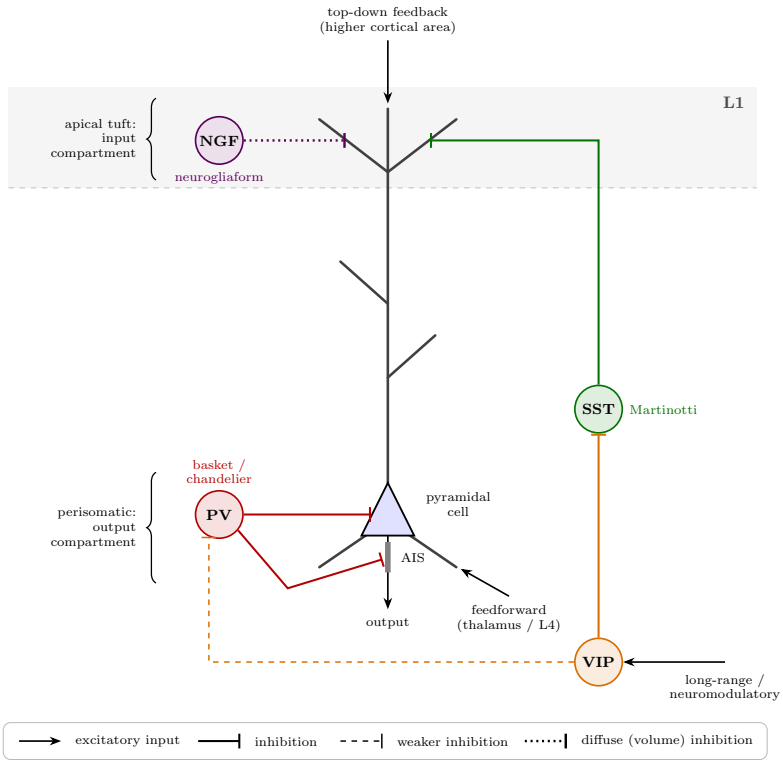


Figure 1.1: Compartment-specific inhibition of a single pyramidal cell. PV cells inhibit the perisomatic output compartment, with basket cells acting at the soma and proximal dendrites and chandelier cells at the axon initial segment (AIS); there they control the gain and timing of spike output. SST Martinotti cells send an ascending axon into layer 1 and inhibit the apical tuft, the input compartment on which top-down feedback arrives. VIP cells inhibit SST cells and, more weakly, PV cells (dashed); when long-range or neuromodulatory input activates VIP cells, the tuft is released from SST inhibition (the VIP $\rightarrow$ SST $\rightarrow$ pyramidal disinhibitory motif). Neurogliaform cells in layer 1 add slow, diffuse inhibition of the tuft by volume transmission (dotted). Feedforward input arrives on the basal dendrites near the soma. Targeting follows Tremblay, Lee, and Rudy [10].

and from layers 2/3 to layer 5, are focused and columnar: each is a dense, vertically restricted axonal projection from one layer onto the next. Layer 6 receives no comparable focused projection from any single layer [8]. The descending axons of layer-2/3 and layer-4 cells pass through it with little branching, and most of the intracortical excitation it receives comes instead from the deep layers themselves, namely layer 5 and layer 6, much of it carried by long, horizontally running axons that link neighbouring columns and areas [8]. The layer-5 contribution is nonetheless genuine. Paired recordings have confirmed direct excitatory connections from layer-5 pyramidal cells, both small adapting cells and large thick-tufted burst-firing cells, onto layer-6 pyramidal cells, and although few pairs have been recorded the connection appears to be common; and locally silencing layer 5 in visual cortex removes the corresponding part of the receptive fields of layer-6 cells [8]. The relative strength of the inputs from different layer-5 cell types onto different layer-6 cell types, however, remains to be characterized.

How layer 5 reaches the corticothalamic output depends on which layer-6 cell is considered. Layer 6 contains two principal pyramidal populations: the corticothalamic (CT) cells that give rise to the modulatory projection of Section 1.6, which make up only some 30–50% of its pyramidal cells, and corticocortical (CC) cells, whose long horizontal axons remain within the cortex [8]. Mapping of inputs by focal glutamate uncaging indicates that CT cells in upper layer 6, the population that projects back to first-order relays, receive excitation from layers 4, 5B, and 6; that CT cells in lower layer 6 receive theirs mainly from within layer 6, with a minor layer-5B component; and that CC cells receive nearly all of theirs from layers 5 and 6 [8]. The directly recorded layer-5-to-layer-6 connections terminated on CC-type cells, and CC cells in turn innervate layer-6 pyramidal cells of all classes, CT cells included, at a high rate [8]. The route from layer 5 to the corticothalamic output therefore runs both directly, onto CT cells, and indirectly, through the layer-6 CC population.

The consequence is that the corticothalamic cell is an integrator rather than one more link in a serial chain. Upper layer-6 CT cells are themselves thalamo-recipient: they receive direct input from the first-order relay to which they project back [8, 12]. They combine this thalamic input with already-processed signals arriving from layer 5 and from layer-6 CC cells [8]. What layer 6 returns to the thalamus (Section 1.6) is therefore not simply the next stage of the column's serial flow, but a combination of the column's processed output with the input the relay itself supplied: a combination suited to adjusting how the relay transmits what follows rather than to carrying a new message. This is why the integrative diagram (Section 1.7) draws the layer-5-to-layer-6 connections as intracortical links outside the numbered driver path.

A feature of the canonical circuit that is easily missed, but is central to how it operates, is that the feedforward afferent is not the numerically dom-

inant input at any stage. The thalamocortical projection accounts for only a small fraction of the excitatory synapses onto its layer-4 target neurons; the overwhelming majority of excitatory synapses in the column originate from other cortical neurons, both within a layer and between layers [4]. The circuit is therefore dominated by recurrent, intracortical excitation rather than by the afferent it relays: the thalamic input is better understood as igniting a recurrent cortical process than as being passively passed along. This recurrent character, a small driving input amplified and shaped by dense local feedback, is the property that makes the column a computational unit rather than a relay, and it is the aspect of the biology most directly relevant to the recurrent architecture the rest of this thesis develops.

The inhibitory interneurons of Section 1.3 are woven into this excitatory template at every stage. In the input layer, thalamocortical afferents contact PV cells as well as excitatory neurons, so that thalamic drive arrives with a brief lag of perisomatic feedforward inhibition that sharpens the timing and gain of the layer-4 response. More generally, because each excitatory projection is accompanied by the compartment-specific inhibition described earlier (perisomatic from PV cells, dendritic from SST cells, disinhibitory gating from VIP cells), the flow along the spine is everywhere shaped by the balance of excitation and inhibition rather than by excitation alone [10].

This canonical picture is an abstraction, and two qualifications on it should be stated rather than absorbed silently. First, the strictly serial reading in which all cortical activity must pass through layer 4 before reaching the deep layers is now known to be incomplete: deep-layer neurons receive sensory-evoked input directly from the thalamus, at the same latency as layer 4 and persisting when layer 4 is silenced, so that the thalamus drives an upper (L4, L2/3) and a lower (L5/6) stratum in parallel rather than purely in series [12]. Second, quantitative and optogenetic studies have refined several of the interlaminar projections of the template, including the targets of the layer-6 projection and the contribution of translaminar inhibition, beyond the resolution of the original abstraction. This chapter adopts the canonical microcircuit as the backbone description of intracolumnar flow because it remains the shared framework the thesis depends on, while treating these refinements as established qualifications to it rather than replacing the backbone with them.

The circuit described so far is intracortical: it ends where the column's output leaves it. But that output does not simply exit: much of it returns to the thalamus, and the thalamus returns signals to the cortex, so that the column is one stage of a larger re-entrant loop. The thalamus, its relay and gating apparatus, and the two descending projections that close this loop are the subject of Sections 1.5 and 1.6.

1.5 The thalamus: relay, drivers and modulators, and gating

Every ascending pathway to the neocortex passes through the thalamus, and much of the cortex's output returns to it, so the thalamus is at once the cortex's obligatory gateway and its re-entrant partner. It is not, however, the passive relay this position might suggest: what it transmits to the cortex is continuously gated and shaped by other inputs, and characterizing that gating is the purpose of this section. Three neuronal populations carry it out. Thalamocortical relay cells are the projection neurons that send the thalamic message to the cortex; local GABAergic interneurons provide inhibition within certain nuclei, though their prevalence varies with nucleus and species; and the thalamic reticular nucleus (TRN), a sheet of GABAergic cells wrapped around the thalamus, provides the principal inhibitory control over the relay cells [7, 2].

The organizing distinction for the entire thalamic circuit, due to Sherman and Guillery [6], is between two kinds of excitatory input: drivers and modulators. A driver is the input that carries the message to be relayed, that is, the one that determines the receptive-field properties of its target cell. Drivers form large synaptic terminals on the proximal dendrites of relay cells, produce large all-or-none excitatory postsynaptic potentials that depress on repeated activation, and act through ionotropic glutamate receptors alone. A modulator, by contrast, does not constitute the message but adjusts how it is transmitted: the gain of transmission and the response mode of the relay cell. Modulators form small terminals on distal dendrites, produce small graded potentials that facilitate on repeated activation, and engage metabotropic as well as ionotropic receptors. A defining and initially counter-intuitive feature of this scheme is that the driver is numerically a minority input: in the lateral geniculate nucleus, for example, the retinal driver afferents form only a small fraction of the synapses onto relay cells, the remainder being modulatory [6, 7]. As in the cortex, then, the input that carries the information is powerful but sparse, while the inputs that set the terms of transmission are numerous.

This distinction furnishes a functional classification of thalamic relays according to the source of their driver. A first-order relay receives its driving input from a subcortical, ascending source and transmits, to the cortex, information that the cortex has not yet seen: the lateral geniculate nucleus, driven by the retina, and the ventral posterior nucleus, driven by the medial lemniscus, are the canonical examples. A higher-order relay instead receives its driving input from the cortex itself, specifically from layer 5 of one cortical area, and relays it onward to another cortical area; the pulvinar, the posterior medial nucleus, and the mediodorsal nucleus are examples [7]. The two therefore occupy different positions in the flow of cortical process-

ing: first-order relays are entry points for new sensory information, whereas higher-order relays carry information between cortical areas by a route that passes through the thalamus. The corticothalamic projections that establish this difference, and the transthalamic pathway it implies, are the subject of Section 1.6.

What the modulatory inputs principally control is the response mode of the relay cell, and with it the gate through which driver information passes. Relay cells operate in two modes, set by the state of a low-threshold, T-type calcium conductance: a tonic mode, in which the cell transmits its driver input faithfully and more or less linearly, and a burst mode, in which the cell responds to driver input with stereotyped high-frequency bursts that no longer track it faithfully [6, 7]. The switch between these modes is one of the main things modulatory inputs accomplish, and the TRN is central to it: its GABAergic cells receive excitatory collaterals both from thalamocortical relay axons and from descending corticothalamic axons as they pass through, and they return inhibition to the relay cells, so that the TRN is positioned to gate the relay it surrounds and, according to the balance of its activation, to switch its firing mode [7, 2]. Thalamic transmission is thus a controlled variable rather than a fixed conductance: the same driver input reaches the cortex or does not, faithfully or as a burst, according to the state imposed on the relay.

The thalamus described here is therefore a gate whose setting is determined largely by its modulatory inputs. Chief among those inputs, and not yet accounted for, are the two descending projections from the cortex itself, one modulatory and one driving, which together make the thalamus not merely an input gateway but a stage within a re-entrant cortico-thalamo-cortical loop. Those projections, the transthalamic route they create, and the closure of the loop are the subject of Section 1.6.

1.6 The cortico-thalamo-cortical loop

The thalamus of Section 1.5 is a gate whose setting is largely determined by descending input from the cortex. That descending input is not of one kind. The cortex projects to the thalamus through two anatomically and physiologically distinct pathways, originating in two different layers and belonging to the two different classes of Section 1.5: a modulatory feedback projection from layer 6, and a driving feedforward projection from layer 5 [6, 7]. The distinction between them is the organizing principle of the whole loop, and the asymmetry in their targets is what separates first-order from higher-order relays.

The layer-6 projection is the corticothalamic feedback proper, and it has all the marks of a modulator. It is numerically massive, with each relay cell contacted by many layer-6 fibers, yet each contact is weak: small terminals

on the distal dendrites of relay cells, producing small excitatory potentials that facilitate on repeated activation and engage metabotropic glutamate receptors [7, 2]. Its collaterals also innervate the corresponding sector of the TRN, so that layer 6 acts on the relay both directly and through the inhibitory nucleus that gates it. Every thalamic relay, first-order and higher-order alike, receives this layer-6 modulation; for a first-order relay it is the sole cortical input. Functionally, layer 6 does not tell the relay what to transmit but sets how it transmits: the gain of the relay and, through its influence on firing mode and on the TRN, whether the driver passes in tonic or burst form.

The layer-5 projection is of the opposite character and has a restricted target. It contacts only higher-order relays, and there it is a driver: large terminals on proximal dendrites, producing large all-or-none potentials that depress on repeated activation and act through ionotropic receptors alone, with a terminal morphology resembling that of a primary sensory afferent [7]. Unlike the layer-6 axons, it does not send collaterals to the TRN on its way. This projection is best understood not as feedback but as the column's output re-entering the thalamus as a fresh driving message: the result of one area's cortical computation, handed to a higher-order relay to be transmitted onward.

The consequence is a route by which cortical areas communicate through the thalamus, in parallel with their direct corticocortical connections. In this transthalamic pathway, layer 5 of one cortical area drives a higher-order relay, which in turn drives layer 4 of a second area; both legs are driver connections, so the message that passes is a genuine information-bearing signal rather than a modulatory adjustment [13, 7]. Because it passes through a thalamic relay, this corticocortical message is subject to exactly the same gating as ascending sensory information, namely TRN inhibition and the switch between tonic and burst mode. The direct corticocortical connections that run alongside it, including the top-down feedback that terminates on the apical tufts in layer 1 (Section 1.2), form the loop's other descending limb; whether these direct projections act as drivers or modulators is not fully resolved and is left open here [6].

These elements compose a single re-entrant circuit, which can now be traced in full. A driving input, subcortical in the case of a first-order relay, reaches the relay cells and, gated by the TRN and by the cell's firing mode, ascends to drive layer 4 of the cortical column. Within the column, the canonical microcircuit of Section 1.4 processes this input through its recurrent laminar flow, combining it with the top-down feedback arriving at the apical tufts. The column then returns two descending signals of the two distinct kinds: layer 6, through corticothalamic cells that have combined the column's processed output with the relay's own input (Section 1.4), sends a modulatory projection back to the relay that supplied its input, and to the matched TRN sector, adjusting the gain and mode of that relay for

what follows; and layer 5 sends a driving projection to a higher-order relay, which transmits the column's output onward to the next area, where the same operations repeat. Ascending drive and descending modulation are thus carried by anatomically and kinetically separate pathways, and every passage through the thalamus, in either direction, is gated. This separation of driving from modulatory signaling, running in both directions around a thalamo-cortical loop, is the central architectural feature of the biology this chapter has described.

The whole of this circuit, comprising its layers and cells, the direction and class of each connection, and the two gated directions of the loop, is summarized in the integrative diagram of Section 1.7.

1.7 Integrative circuit diagram

Figure 1.2 brings the elements of this chapter together in a single diagram. It traces the driving signal through the thalamus and the column as a numbered sequence, distinguishes driving connections from modulatory ones and from the inhibitory gate of the TRN, and shows layer 6 resolved into its corticothalamic and corticocortical populations (Section 1.4). The figure is the reference circuit on which the rest of the thesis draws.

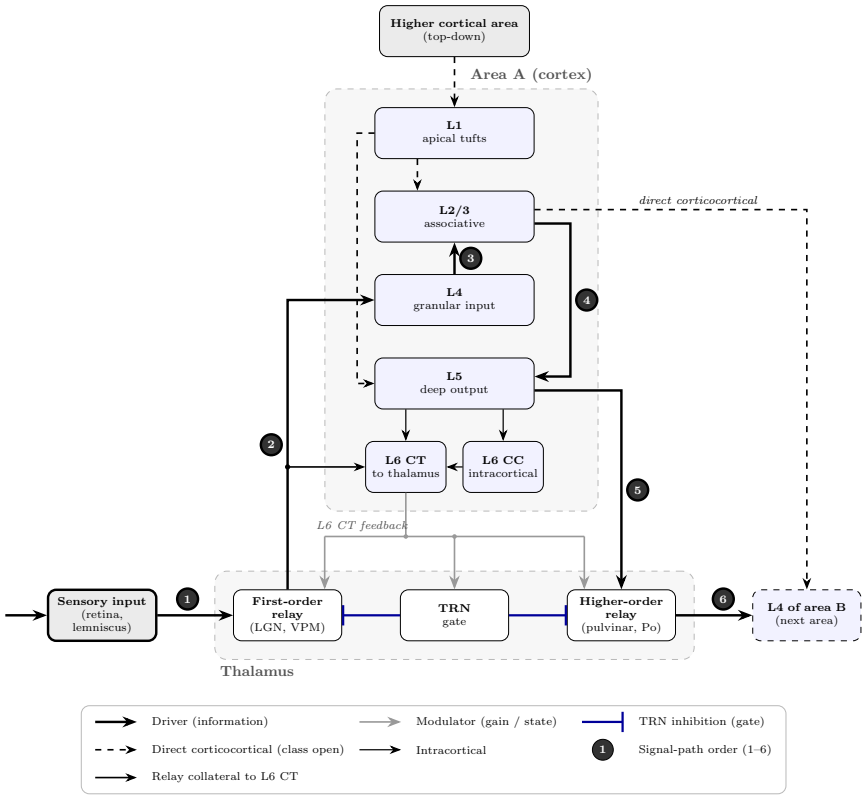


Figure 1.2: The cortico-thalamo-cortical loop and the canonical cortical column, traced as a numbered flow. (1) Sensory input reaches a first-order relay, which (2) drives layer 4. (3–4) The column processes it through the canonical intracolumnar flow (L4 → L2/3 → L5). (5–6) Layer 5 sends a driving projection to a higher-order relay, which drives layer 4 of the next cortical area (the transthalamic route). Layer 6 is drawn as its two pyramidal populations. Layer 5 projects to both corticothalamic (CT) and corticocortical (CC) cells, and CC cells in turn excite CT cells (thin, intracortical). CT cells also receive a collateral of the first-order relay's thalamocortical axon, so that they combine the column's processed output with the relay's own input. Only the CT cells return a modulatory projection (thin grey) to both relays and the TRN, and the TRN gates both relays (blue, bar-ended). Direct corticocortical projections (dashed, class open) run alongside the transthalamic route: feedforward from L2/3 to layer 4 of the next area, and top-down feedback from a higher cortical area onto the apical tufts of L2/3 and L5 pyramidal cells in layer 1. The relays' collateral input to the TRN and the long horizontal axons of CC cells are omitted for clarity. Connection classes follow Sherman and Guillery [6, 7]; the organization of layer 6 follows Thomson [8].

Chapter 2

Computational Foundation

2.1 Scope and Review Axes

Chapter 1 described the circuit: the laminar column, its excitatory and inhibitory cell classes, and the cortico-thalamo-cortical loop. This chapter reviews the computational side. It investigates how existing models represent cortical inhibition, sparse coding and associative memory, and what they have accomplished. The aim is not a catalogue. Each model is read for what it establishes and also for where it stops, so that the boundary of current work is visible when the gap is stated.

Three axes structure the review. The first is the *substrate*: the medium in which a model represents and computes. Candidates include binary sparse distributed representations over minicolumns with dendritic segments, rate-coded populations, spiking attractor networks, and multi-compartment neurons. The substrate matters because a mechanism shown in one medium does not automatically carry over to another. The second axis is the *inhibition model*. Inhibition is either homogeneous, a single mechanism such as k-winners-take-all applied uniformly, or differentiated, with distinct interneuron classes that differ in target compartment, dynamics and arithmetic effect. The third axis is *encoding versus retrieval*. It asks whether a model separates a mode in which new associations are stored from a mode in which stored associations are recalled, and what controls the switch.

The third axis has a clear precedent outside neocortex. In the hippocampal model of Hasselmo et al. [14], the theta rhythm alternates between two phases. In one, entorhinal input is strong and long-term potentiation is strong, but recurrent input from CA3 is weak, so new associations are encoded without interference from old ones. In the other, the balance reverses and stored associations are retrieved. That separation is implemented by oscillatory and cholinergic modulation in hippocampus. Whether the interneuron circuitry of a neocortical column could implement a comparable arbitration is the question this review keeps in view.

The chapter proceeds as follows. Section 2.2 reviews the HTM tradition and its single model of inhibition. Section 2.3 covers associative extensions of that tradition, Sparse Distributed Memory and the Neural Association Algorithm. Section 2.4 reviews circuit models that assign distinct roles to PV, SST and VIP interneurons. Section 2.5 reviews associative-memory columns in the attractor and BCPNN tradition. Section 2.6 examines the two operator mechanisms at issue, divisive normalisation and dendritic arbitration. Section 2.7 places all of these models on the three axes.

2.2 The HTM, CLA and SDR Tradition

Hierarchical Temporal Memory (HTM) and its Cortical Learning Algorithms (CLA) form a family of models built on a small set of neocortical principles: sparse distributed representations, minicolumns, pyramidal neurons with active dendrites, and continuous local learning [15]. The tradition matters to this thesis because it provides the substrate that the thesis builds on. It is equally significant for what it leaves out.

2.2.1 Sparse Distributed Representations

A sparse distributed representation (SDR) is a large binary vector in which only a small fraction of bits are active at any time, typically a few percent. Ahmad and Hawkins [16] derived the properties that make SDRs suitable for cortical computation, concerning their scaling, robustness and generalisation. The overlap between two SDRs measures their similarity. The probability that two unrelated SDRs share many active bits falls rapidly as dimensionality grows, so a pattern can be recognised reliably by sampling only a small subset of its active bits. The same property lets a set of SDRs be stored as their union and still be matched with a low false-positive rate. Ahmad and Hawkins [17] later linked these properties to dendritic segments: a segment that detects a coincidence among a small sample of active inputs behaves as a robust pattern detector over SDRs.

2.2.2 The Spatial Pooler

The Spatial Pooler (SP) converts arbitrary binary input into an SDR over a set of minicolumns [18]. Each minicolumn has a pool of potential feedforward synapses, each with a scalar permanence. A synapse counts as connected when its permanence exceeds a threshold, and the minicolumn's overlap is the number of connected synapses whose inputs are active. A minicolumn becomes active if its overlap exceeds a stimulus threshold and ranks within the top s percent of its neighbourhood, with s typically 2%. The authors describe this selection as a k-winners-take-all computation. When the neighbourhood covers the whole layer, it becomes global inhibition. Learning

is Hebbian: winning minicolumns strengthen synapses from active inputs and weaken synapses from inactive ones. A slower homeostatic mechanism, boosting, raises the excitability of minicolumns that are active too rarely. Cui et al. showed that the Spatial Pooler adapts quickly to changing input statistics, becomes more robust to noise with learning, uses its minicolumns efficiently, and degrades gracefully under cell death.

Dobric et al. [19, 20] found that the same homeostatic mechanism can destabilise learned representations, pushing the Spatial Pooler into unstable, "epileptic" activity. Their remedy confines homeostatic plasticity to an initial "newborn stage" and then disables it, after which learned SDRs remain stable.

2.2.3 Temporal Memory and Active Dendrites

Hawkins and Ahmad [21] extended the HTM neuron with three integration zones. The proximal zone receives feedforward input and drives action potentials. The basal zone receives contextual input, mostly from nearby cells in the same region. The apical zone receives feedback. Each dendritic segment acts as an independent pattern detector, so a single neuron can recognise hundreds of distinct patterns. A match on a basal or apical segment does not fire the cell. It depolarises the cell slightly, putting it in a predictive state.

The Temporal Memory (TM) network uses this predictive state for sequence learning. All cells in a minicolumn share the feedforward receptive field selected by the Spatial Pooler. When a minicolumn becomes active, its depolarised cells fire first and inhibit their neighbours, so the input is represented by a sparse, context-specific subset of cells. If no cell in the minicolumn was predicted, all of its cells become active, which signals an unanticipated input. Because predictions are sparse, the network can hold many simultaneous predictions without confusion. Cui et al. [22] showed that Temporal Memory learns variable-order sequences, keeps multiple predictions open until disambiguating input arrives, and matches the prediction accuracy of ARIMA (autoregressive integrated moving average) models, extreme learning machines, LSTM and echo-state networks on artificial and real streaming data.

2.2.4 Continual Unsupervised Learning

Learning in this lineage is local, Hebbian and continuous. There is no global error signal and no separation between training and testing. Cui et al. [22] reported continuous online learning, robustness to sensor noise, fault tolerance, and good performance without task-specific hyperparameter tuning. Iyer et al. [23] carried the dendritic principle into deep networks for continual learning. They gated units through context-modulated dendritic segments

and enforced k-winners-take-all sparsity, and the combination reduced catastrophic forgetting across sequentially learned tasks.

The same property has a consequence for the third review axis. Because learning is always on, an HTM network stores and recalls in a single pass. It has no state in which input is recalled without being learned, and no state in which input is learned without being matched against what is already stored. The lineage contains no mechanism comparable to the encoding and retrieval phases described in Section 2.1.

2.2.5 Inhibition in the HTM Lineage

Across the Spatial Pooler, Temporal Memory and their continual-learning extensions, inhibition is a single operation: rank the candidates and keep the top k, applied either globally or within a neighbourhood. It has no cell identity, no target compartment and no dynamics. Its arithmetic effect is fixed: it caps the active population at k and depends only on the ranking of inputs, not on how strong they are. The only route to a smaller population is the separate stimulus threshold, which is not part of the inhibition. Inhibition also does not act separately on the apical zone, even though the HTM neuron has one, so it offers no means of weighing feedback against feedforward drive.

This is a modelling choice, not a claim about biology, and the authors' own discussions point to more structure than the algorithms use. Hawkins and Ahmad [21] propose fast-spiking basket cells as the most likely source of the rapid inhibition that TM depends on. These are the parvalbumin-expressing perisomatic cells described in Section 1.3. Cui et al. [18] note that the cortex contains many distinct classes of inhibitory neurons and suggest that classes innervating cells within a minicolumn, such as bipolar and double bouquet cells, may help define it. The lineage therefore attributes its inhibition to specific and differing cell types in biology, but implements a single homogeneous rule in computation.

Two questions follow. First, whether replacing k-winners-take-all with a PV-like divisive operator changes behaviour that kWTA fixes by construction, namely assembly size and the possibility of complete suppression. Second, whether adding an SST-like dendritic operator gives the network a handle on feedback that it currently lacks. Section 2.6 examines both mechanisms. On the three axes, the HTM tradition provides the substrate, a homogeneous inhibition model, and no separation of encoding from retrieval.

2.3 Associative Extensions

The HTM algorithms reviewed above encode inputs and learn sequences within a single region. Two lines of work take sparse binary representations further, from representation to association. Sparse Distributed Memory is

an earlier and independent formalism for storing and recalling patterns in a high-dimensional binary space. The Neural Association Algorithm extends the HTM tradition itself to associations across cortical areas.

2.3.1 Sparse Distributed Memory

Kanerva's Sparse Distributed Memory (SDM) stores binary words of very high dimension, typically around a thousand bits [24]. The address space is far too large for every address to have physical storage, so the memory is built from a fixed random sample of addresses, called hard locations, each holding one counter per bit. A word is written by activating every hard location within a given Hamming radius of its address and adding the word to their counters, incrementing for ones and decrementing for zeros. A word is read by activating the hard locations around the cue address, summing their counters bit by bit, and thresholding the sums. Because each word is spread across many locations and each location holds traces of many words, reading is tolerant to noisy cues. Reading repeatedly, with each output used as the next address, converges to the stored word when the cue lies close enough to it. Storing each item at the address of its predecessor turns the same mechanism into a sequence memory. Kanerva related the architecture to the circuitry of the cerebellum.

SDM matters to this review in three ways. First, its hard locations play a role similar to the minicolumns of the Spatial Pooler: both map an input onto a sparse set of units chosen by similarity. They differ in how that set is chosen. The SDM addresses are fixed and random rather than learned, and the active set is every location inside a radius, so its size varies with the cue. The Spatial Pooler learns its receptive fields but caps the active set at exactly k. SDM therefore needs no inhibition to stay sparse, and its active population is free to grow or shrink, which is the property that kWTA removes (Section 2.2.5). Second, SDM separates writing from reading cleanly, as two distinct operations. That separation is imposed from outside, however: the caller decides whether a cycle writes or reads, and nothing in the memory makes that decision. Third, SDM is an abstract memory and not a model of cortical tissue. It has no minicolumns, dendrites or laminae, so it says nothing about how a cortical circuit might implement either its selection rule or its write and read modes.

A note on terminology is needed here. The title of this thesis describes its model as a laminar sparse-distributed memory column. The term is used in its descriptive sense: a cortical column that stores and recalls associations over sparse distributed representations. The column is built on the HTM substrate of Section 2.2, not on Kanerva's architecture of random hard locations and counters. The two share the principle that makes such memories robust, namely storage distributed across many sparsely active units. Kanerva's model is reviewed here because it is the original statement of that

principle, and because it makes explicit two properties that the HTM lineage lacks: an active set whose size varies with the input, and a separation of writing from reading.

2.3.2 The Neural Association Algorithm

The Neural Association Algorithm (NAA) extends HTM from a single region to a cortical space of many areas, each running the same algorithm [25]. In each area an encoder, usually the Spatial Pooler, produces an SDR. An association function then maps the active cells of any area onto a sparse set of active cells in a target area. NAA uses the three segment types of the HTM neuron for this. Proximal segments connect to sensory input, distal (basal) segments connect to cells of other minicolumns in the same area, and apical segments connect to cells in other areas. Each segment is active or matching, depending on how many connected synapses it has relative to two thresholds. NAA distinguishes two kinds of context. A spatial context is an undirected association between two SDRs, such as a song and the place where it was heard. A temporal context, by contrast, is a directed association, created when one state follows another. Learning happens only inside active minicolumns. There, synapses from active cells onto active segments are strengthened and new synapses are grown, while synapses onto matching segments in inactive minicolumns are weakened. Alongside the association model, the work introduces the newborn stage of the Spatial Pooler (Section 2.2.2) and a parallel implementation of minicolumn computation based on the actor programming model.

NAA is the closest predecessor of this thesis. It shares the substrate and aims at association, which an SDR column must support before encoding and retrieval can be told apart. Its scope on the other two axes is stated openly by its author. On inhibition, the implementation "omits using explicit inhibitory cells for performance reasons" and replaces them with an inhibition algorithm [25, Sec. 5.7.3]. That algorithm comes in a global and a local version, both selecting the minicolumns with the highest overlap at a fixed density, which is the k-winners-take-all rule of Section 2.2.5. No significant difference in learning performance was found between the versions, and the thesis names the investigation of inhibition as future work, specifically how the inhibition process influences sparsity and the overall performance of learning [25, Sec. 9.5.1.7]. On encoding versus retrieval, NAA learns in every cycle, as HTM does. Recalling an input from its SDR, which the thesis calls reverse encoding, is currently handled by a classifier component that is not aligned with biology, and replacing it with learned associations is again listed as future work [25, Sec. 9.5.1.2]. Finally, only the single-area core of NAA is implemented. The apical segments that would link areas, and with them any channel for feedback, are part of the model but not yet of the implementation [25, Sec. 9.5.1.1].

On the three axes, NAA provides the SDR substrate extended toward association, a homogeneous inhibition model that its author explicitly leaves open, and no circuit-level separation of encoding from retrieval. These open items are deliberate simplifications of a framework built for scale, not oversights. Section 2.7 returns to them as the point at which this thesis begins.

2.4 Interneuron-Resolved Circuit Models

Alongside the HTM tradition, a separate body of work models cortical inhibition at the level of interneuron classes. These models take the connectivity described in Section 1.3 as their starting point. PV cells inhibit the perisomatic region of pyramidal cells, SST cells inhibit their dendrites, and VIP cells inhibit SST cells and so release the dendrites from inhibition. The models then ask what this division of labour computes. They are reviewed here in four groups: disinhibitory gating, state-dependent responses to modulation, learning and optimisation within the interneuron circuit, and large-scale laminar models.

Disinhibitory gating. Yang et al. [26] proposed that the SST and VIP motif gates input pathways on the dendrites of pyramidal cells. Their pyramidal neuron has a spiking soma and several independent dendritic compartments, reduced for network simulations to a multi-compartment rate model. SST cells keep every dendritic branch inhibited by default, and top-down control signals onto VIP cells release selected branches. If the inputs of one pathway cluster on the same branches, disinhibiting those branches gates that pathway on while the others stay closed. This selectivity survives dense and even random interneuron connectivity. A calcium-based plasticity rule regulated by dendritic inhibition produces the required clustering, and the gated circuit performs a context-dependent decision task in which a rule signal selects whether motion or colour determines the response.

Hertäg and Sprekeler [27] studied how the same motif redistributes inhibition between the soma and the dendrite. Their rate model uses a two-compartment pyramidal cell, with PV cells inhibiting the soma, SST cells inhibiting the apical dendrite, and VIP cells inhibiting only SST cells. Mutual inhibition between VIP and SST cells turns weak inputs to VIP cells into large shifts between somatic and dendritic inhibition. When that mutual inhibition is strong, the circuit becomes bistable: a transient input switches it persistently between two processing modes, one in which top-down input to the apical dendrite is integrated and one in which it is cancelled. Of the models in this section, this is the closest analogue to the arbitration that this thesis is concerned with, a circuit-level switch that decides how feedback is treated. It is framed as a mode of processing top-down input, however, not as a choice between storing and recalling a memory, and the model learns

no associations.

State-dependent responses. Garcia del Molino et al. [28] built a rate model of pyramidal, PV, SST and VIP populations to explain a puzzling observation from mouse visual cortex. The response of SST cells to top-down behavioural modulation, which targets VIP cells, can change sign when the visual input changes. Their analysis traces this response reversal to specific connection strengths and shows that the effect of activating one interneuron class depends on the state of the whole network. For this thesis the point is a cautionary one. A single interneuron class cannot be assigned one fixed effect from its connectivity alone, so any abstraction of PV or SST as one operator is a simplification whose limits must be stated.

Learning and optimisation. Two models show that the differentiation itself can be learned or derived. Hertäg and Sprekeler [29] added inhibitory plasticity to a circuit of pyramidal, PV, SST and VIP cells and showed that it learns to produce prediction-error neurons, which signal mismatches between sensory input and a prediction. Keijser and Sprekeler [30] took a normative route. They optimised the connectivity and short-term plasticity of a spiking interneuron circuit for compartment-specific feedback inhibition, and the interneurons diversified during optimisation into two classes resembling PV and SST cells. The resulting circuit acts as a decoder that inverts the nonlinear computation within the pyramidal cell. This gives computational support for treating the PV and SST split as a functional one: two classes emerge when inhibition must be specific to compartments.

Laminar models. At the largest scale, Billeh et al. [31] integrated anatomical and physiological data into models of mouse primary visual cortex spanning all layers, with about 230,000 neurons in 17 cell classes. The model exists in a biophysically detailed version and a point-neuron version with identical connectivity, and both were compared against recordings from awake mice. Moreni et al. [32] built a smaller model at the scale of a single column, drawing its connectivity from the same data. Their network of 5,000 leaky integrate-and-fire neurons has pyramidal, PV, SST and VIP populations in layers 2/3, 4, 5 and 6, and VIP cells alone in layer 1. They compared two input regimes. Feedforward input to layer 4 produced net excitation of the column, whereas feedback-like input to layer 5 produced net inhibition, which the authors attribute to translaminar gain control through column-wide inhibition from layer 6 (see Section 1.4). The response to perturbing any single cell group also depended on the state of the column: strong feedforward input made the column less sensitive to every perturbation, while feedback input modulated interactions within it. How the column treats its inputs therefore depends on whether it is driven from below or from above,

which bears directly on the arbitration question of this thesis. Like Billeh et al., however, the model has no synaptic plasticity and performs no memory task. Laminar models of this kind show that interneuron detail across layers can be assembled and constrained by data. Their question is how the circuit responds in different states, not what it stores or retrieves.

Where these models stop. On the inhibition axis, these models provide what the HTM tradition lacks: distinct interneuron classes with distinct targets and dynamics, and analyses of what each contributes. On the other two axes they stop short. Their substrate is rate-coded or spiking populations, with no sparse distributed representations, no minicolumns and no learned assemblies. Their questions are gating, modulation, error signalling and physiological fidelity. None of them stores a set of patterns and then recalls one from a partial cue, so none can ask whether PV and SST contribute differently to encoding and to retrieval. The models supply the mechanisms that this thesis transfers, but not the memory task or the substrate in which it is tested.

2.5 Associative-Memory Columns

A third line of work asks directly whether networks of realistic cortical neurons can act as associative memories. It descends from Hebb's cell-assembly theory and its formal counterpart, the attractor network, in which a stored pattern is a stable state that the network falls into from a partial or noisy cue. Since the 1990s, Lansner and colleagues have tested this idea with increasingly detailed models of cortical layers 2/3, many of them built around a Bayesian-Hebbian learning rule known as BCPNN (Bayesian Confidence Propagation Neural Network) [33]. Because these models store and recall patterns, they are the only family reviewed so far that addresses the memory task directly. Their treatment of inhibition has also grown more differentiated over time.

Columns as memory units. Fransén and Lansner [34] modelled the horizontal network of layers 2/3 as a set of connected cortical columns, with the column rather than the single neuron as the functional unit. Each column contained biophysically detailed pyramidal cells and fast-spiking inhibitory interneurons representing basket cells. A memory was an assembly of columns linked by sparse, long-range excitatory connections. Connection strengths were not learned during the simulation. They were computed in advance with the Bayesian rule from the full set of patterns to be stored. Positive weights became excitatory connections between the pyramidal cells of different columns. Negative weights were routed onto the basket cells of the receiving column, so that inhibition between competing assemblies was

disynaptic. With this wiring the network completed an assembly from a partial cue, resolved rivalry between overlapping assemblies, and sustained activity after the cue only when the excitability of its pyramidal cells was raised by simulated neuromodulation. The model thus showed that biophysically realistic neurons can support attractor memory. It used a single class of interneuron, and the switch between transient and sustained activity came from a neuromodulator, not from the inhibitory circuit.

Modular attractor networks. Later versions organised the same layer 2/3 network into hypercolumns, each containing a set of minicolumns [35]. Basket cells within a hypercolumn provide lateral inhibition that makes each module behave as a soft winner-take-all, typically allowing one active minicolumn at a time, and a stored pattern is a set of minicolumns distributed across hypercolumns. Regular-spiking non-pyramidal cells, possibly double bouquet cells, provided inhibition between minicolumns [36]. This organisation is strikingly close to the minicolumn and k-winners-take-all structure of the Spatial Pooler (Section 2.2.2). The difference is that the competition is implemented by a population of basket cells rather than by a sorting rule.

Differentiated interneurons in memory columns. Three studies added further interneuron classes to this architecture. Krishnamurthy, Silberberg and Lansner [36] introduced Martinotti cells, which receive strongly facilitating synapses from pyramidal cells, alongside basket cells, whose input synapses depress. When the network entered a stored attractor, basket cells fired first and then tapered off, while Martinotti cells engaged with a delay and shifted the balance of excitation and inhibition until the attractor terminated. A sensitivity analysis showed that Martinotti cells, rather than cellular adaptation or synaptic depression, dominated the attractor dwell time, and replacing them with additional basket cells did not reproduce the effect. The temporal profile of each interneuron class, not only the amount of inhibition, therefore shaped the memory dynamics. In a follow-up model, long-range recruitment of Martinotti cells produced surround suppression and promoted the saliency of non-uniform stimuli, and disinhibition through VIP cells removed that suppression [37]. Chrysanthidis, Fiebig and Lansner [38] addressed a different problem. In a spiking BCPNN model with hypercolumns and minicolumns, the learning rule produces negative as well as positive weights, which a single pyramidal cell cannot express under Dale's principle. Adding double bouquet cells, whose axons stay within a minicolumn, turned the learned negative weights into a plastic disynaptic pathway. Basket cells continued to provide non-specific feedback inhibition, and the network's learning and memory function were preserved.

Where these models stop. This tradition comes closest to the present thesis on two of the three axes. It studies associative memory directly, and it has already shown that differentiated interneurons matter inside a memory network. The roles it assigns them are specific, however: basket cells implement competition between minicolumns, Martinotti cells control how long an attractor lasts and suppress its surround, and double bouquet cells make learned inhibition compatible with Dale's principle. None of these models asks whether one interneuron class governs the storage of new patterns and another their recall. Where the mode of the network does change, as in the sustained activity of Fransén and Lansner, the switch is neuromodulatory. The substrate also differs. These are spiking networks whose memories are assemblies of minicolumns selected by soft winner-take-all competition, not learned SDRs over dendritic segments, and in the earliest models the stored patterns were wired in rather than learned. What interneuron-differentiated inhibition would add to this tradition is therefore not the presence of several interneuron classes, which it already has, but a test of whether PV-like and SST-like inhibition divide the work of encoding and retrieval between them.

2.6 Two Operator Mechanisms

The preceding sections point to two mechanisms that the HTM tradition lacks and that the interneuron literature associates with PV and SST cells: a divisive control of overall response magnitude, and a dendritic control of how feedback arriving at the apical tuft is combined with feedforward drive. This section reviews each as a computational operator, states the evidence that links it to an interneuron class, and states where that link is contested.

2.6.1 Divisive Normalisation

Carandini and Heeger [39] proposed normalisation as a canonical neural computation. In its standard form, the response of a neuron is its driving input divided by a constant plus the summed activity of a pool of neurons,

$$R_j = \gamma \, \frac{D_j^n}{\sigma^n + \sum_k D_k^n}, \tag{2.1}$$

where D_j is the drive to neuron j, the sum runs over the normalisation pool, σ sets the input level at which normalisation takes effect, and γ and n set the gain and the steepness of the response. First developed to explain responses in primary visual cortex, normalisation has since been invoked across sensory modalities and for operations such as the effects of attention, the encoding of value and multisensory integration. Carandini and Heeger stress that a canonical computation need not have a single implementation: different circuits, regions or species may compute it with different components.

For this thesis the contrast with k-winners-take-all is the important point. kWTA acts on the rank of inputs and fixes the number of active units at k. Normalisation acts on their magnitude: each response is scaled relative to the activity of the pool, and the ordering of inputs is preserved. Combined with a firing threshold, the number of units that remain active is then free to depend on the input, and when the pool is strongly driven every unit can in principle fall below threshold. Variable assembly size and reachable complete suppression are exactly the two properties that kWTA excludes by construction (Section 2.2.5), and whether a PV-like normalising operator restores them in an SDR column is the first question this thesis tests.

The attribution of division to PV cells, however, is not settled. Wilson et al. [40] activated and suppressed PV and somatostatin-expressing (SOM, here SST) cells optogenetically in mouse visual cortex and reported a clean dissociation: PV cells mainly divided responses, changing their gain while preserving stimulus selectivity, whereas SST cells mainly subtracted from them, shifting response levels and sharpening selectivity. Studies published at the same time drew different conclusions. Lee et al. [41] found that activating PV cells sharpened orientation and direction tuning and improved orientation discrimination in behaving mice, while activating SST or VIP cells had no significant effect. Atallah et al. [42] found that PV activity transformed responses linearly, subtracting and scaling them with little change in tuning width. El-Boustani and Sur [43] then showed that the operation performed by SST cells depends on the stimulus: for small stimuli SST cells respond later than their targets and inhibit subtractively, but for large stimuli they respond at the same time and inhibit divisively. They concluded that the arithmetic of inhibition is a dynamic property of the circuit, not a fixed property of a cell class. Seybold et al. [11] reached a similar conclusion from a different direction, showing that recurrent network integration can turn subtractive inhibition of single neurons into divisive inhibition of the population, or the reverse.

This evidence fixes how the operators are used in this thesis. As stated in Section 1.3, the synaptic targeting of PV and SST cells is robust, while the assignment of a fixed arithmetic operation to each class is not. The thesis therefore treats "PV as divisive normalisation" and "SST as dendritic gating" as operator hypotheses: modelling choices motivated by targeting and by the studies above, whose consequences are tested rather than assumed. This is also why the comparison with kWTA is framed as a falsification check. If a PV-like normalising operator in an SDR column behaves no differently from kWTA, the operator distinction has no computational consequence at this level of abstraction, and the thesis reports that result.

2.6.2 Dendritic and Apical Arbitration

The second operator concerns where inhibition acts on the pyramidal cell rather than how strongly. Larkum, Zhu and Sakmann [44] showed that layer 5 pyramidal neurons have a second spike-initiation zone in the distal apical dendrite. Input to the apical tuft alone is strongly attenuated before it reaches the axon. If a back-propagating action potential from the soma coincides with distal input within a window of a few milliseconds, however, the dendrite fires a calcium spike and the neuron emits a burst. Inhibition of the distal dendrite can block the calcium spike selectively and so prevent the burst. Larkum [9] later proposed this coupling as an organising principle of cortex. Feedforward input arrives near the soma, feedback from other areas arrives at the apical tuft in layer 1, and the pyramidal cell itself acts as an associative element that signals when the two coincide. In this view the inhibitory control of the dendrite decides whether feedback is allowed to combine with feedforward drive. Murayama et al. [45] identified the interneurons responsible in vivo. Martinotti cells in deep layers, a class of SST interneuron, inhibit the apical tufts of layer 5 neurons and set the slope and threshold of their dendritic calcium response to sensory stimulation, and disynaptic inhibition through these cells can abolish dendritic calcium spikes entirely.

Two models have turned dendritic arbitration into a learning principle. Urbanczik and Senn [46] proposed a plasticity rule in which dendritic synapses change so as to reduce the mismatch between the dendritic potential and the somatic firing of the same neuron. Sacramento et al. [47] built on this rule in a multilayer network of multi-compartment pyramidal neurons. In their model, lateral interneurons that they identify with SST cells learn to predict the top-down feedback arriving at the apical dendrite and to cancel it. Whatever feedback is not cancelled is a local prediction error, and plasticity driven by that error approximates the backpropagation algorithm without separate learning phases. Together with the disinhibitory gating of Yang et al. [26] (Section 2.4), these models establish dendrite-targeting inhibition as the natural candidate for deciding how much feedback a pyramidal cell integrates.

Where these mechanisms stop. Both operators are well developed, but neither has been placed in the setting of this thesis. Normalisation is defined independently of any circuit, its attribution to PV cells is contested, and it has not been tested as a replacement for kWTA in an SDR column. Apical arbitration is established at the level of single cells and of circuits for gating and learning, but the questions asked of it are coincidence detection, pathway selection and credit assignment in supervised learning. In Sacramento et al. the SST-like interneurons cancel predictable feedback so that errors can drive learning. They do not decide whether the network stores a new pattern

or recalls a stored one, and the model's explicit aim of avoiding separate phases is the opposite of the encoding and retrieval separation examined here. Whether a dendritic operator of this kind, placed in an SDR column alongside a normalising one, can arbitrate between feedforward encoding and feedback-driven retrieval remains open.

2.7 Synthesis: Relation to Prior Work

Table 2.1 places the models reviewed in this chapter on the three axes introduced in Section 2.1. Read by rows, it shows that each tradition is strong on one or two axes and silent on the rest. Read by columns, it shows where the empty cell lies.

The HTM tradition and its associative extension supply the substrate. Sparse distributed representations over minicolumns with dendritic segments, learned continuously by local rules, are what the present model is built from. Their inhibition, however, is a single homogeneous k-winners-take-all rule, which the authors of the lineage themselves describe as a simplification (Sections 2.2.5 and 2.3.2), and learning is always on, so encoding and retrieval are never separated. The interneuron-resolved circuit models supply the opposite. They give PV, SST and VIP cells distinct targets, dynamics and roles, but in rate-coded or spiking populations without sparse distributed representations, and in tasks of gating, modulation, error signalling and physiological fidelity rather than memory. The attractor and BCPNN tradition comes closest from a third direction. It studies associative memory in cortical columns and already includes several interneuron classes, but the roles it assigns them are competition, attractor dwell time, surround suppression and compliance with Dale's principle. The two operator mechanisms, divisive normalisation and dendritic arbitration, are each well established, yet neither has been tested as a replacement for kWTA in an SDR column, and the assignment of each to a single interneuron class remains contested.

The encoding and retrieval column of the table is the most telling. Where a model separates the two at all, the switch lies outside the inhibitory circuit. In Sparse Distributed Memory the caller decides whether a cycle writes or reads. In the hippocampal model of Hasselmo et al. and in the cortical model of Fransén and Lansner the change of mode comes from neuromodulation. The one interneuron model with a persistent switch between modes, that of Hertäg and Sprekeler, applies it to the processing of feedback, not to the storage and recall of patterns. Sacramento et al. go further in the other direction and design their circuit so that no separate phases are needed.

The gap that follows from this review has three parts, and no existing model combines them: a discrete SDR substrate in the HTM tradition; inhibition resolved into two operator-distinct interneuron types, a PV-like divisive normalisation and an SST-like dendritic gate; and an explicit test

of whether these two operators contribute differently to encoding and to retrieval. The contribution claimed here is correspondingly narrow. It is not a new account of cortical inhibition, but the transfer of an established biological distinction into a substrate that has so far lacked it, together with a test of whether the transfer has a functional consequence that homogeneous kWTA cannot produce.

Model	Substrate	Inhibition model	Question tested	Encoding vs. retrieval
HTM, CLA and SDR tradition (Section 2.2)				
Spatial Pooler, Temporal Memory [18, 21]	Binary SDRs, minicolumns, dendritic segments	Homogeneous kWTA, global or local	Sparse coding, sequence learning	Not separated
Iyer et al. [23]	Deep network with dendritic segments	Homogeneous kWTA	Continual multi-task learning	Not separated
Associative extensions (Section 2.3)				
SDM [24]	Abstract binary memory	None; radius selection	Storage and noisy recall	Separate, set by caller
NAA [25]	HTM-derived SDRs, multiple areas	kWTA algorithm, no inhibitory cells	Association, SP stability	Not separated
Interneuron-resolved circuit models (Section 2.4)				
Yang et al. [26]	Rate, multi-compartment pyramidal cell	PV, SST, VIP	Pathway-specific gating	Not addressed
Hertäg and Sprekeler [27]	Rate, two-compartment pyramidal cell	PV, SST, VIP	Somato-dendritic redistribution	Mode switch for feedback only
Garcia del Molino et al. [28]	Rate populations	PV, SST, VIP	Response reversal	Not addressed
Hertäg and Sprekeler [29]	Rate, plastic interneurons	PV, SST, VIP	Prediction-error learning	Not addressed
Keijser and Sprekeler [30]	Spiking, optimised circuit	Emergent PV- and SST-like classes	Compartment-specific inhibition	Not addressed
Billeh et al. [31]	Spiking laminar V1, 230,000 cells	17 data-driven cell classes	Physiological fidelity	Not addressed
Moreni et al. [32]	Spiking laminar column	PV, SST, VIP per layer	Feedforward vs. feedback states	Not addressed
Associative-memory columns (Section 2.5)				
Fransén and Lansner [34]	Biophysical spiking columns	Basket cells	Completion, rivalry	Retrieval only; neuro-modulatory mode
Lundqvist et al. [35]	Spiking hyper- and minicolumns	Basket soft WTA, RSNP cells	Attractor dynamics	Retrieval only
Krishnamurthy et al. [36, 37]	Spiking attractor network	Basket, Martinotti (VIP in 2015)	Dwell time, surround suppression	Retrieval only
Chrysanthidis et al. [38]	Spiking BCPNN network	Basket, double bouquet	Learned inhibition, Dale's principle	Not arbitrated by interneurons
Operator mechanisms (Section 2.6)				
Normalisation [39]	Circuit-independent	Divisive	Canonical computation	Not addressed
Sacramento et al. [47]	Rate, multi-compartment network	SST-like apical prediction	Supervised credit assignment	Single phase by design
This thesis	**SDR laminar column**	**PV-like divisive, SST-like dendritic**	**Encoding vs. retrieval dissociation**	**Tested**

Table 2.1: Relation of this thesis to prior work on the three review axes: substrate, inhibition model, and whether encoding and retrieval are separated.

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